# Kayané ROBACH

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## RESEARCH INTEREST

*Causal Inference, Data Fusion, Bayesian Statistics, Time-to-Event, Explainability, Uncertainty Quantification*

## EDUCATION

**2022 – 2026 PhD in Mathematical Statistics; Amsterdam UMC, The Netherlands**

**Causal Record Linkage** *Development of new statistical methods to estimate causal effects based on heterogeneous data sources.*

*Supervisors: Stéphanie van der Pas, Michel Hof, Mark van de Wiel*

**2021 – 2022 Master 2 Mathematics and Applications; ENS Paris Saclay, France**

**MVA (Mathematics, Vision, Learning)** *Topological data analysis, Object recognition and computer vision, Responsible Machine Learning, Computational statistics, Geometry of shape space, Probabilistic graphical models, Advanced learning for text and graph data, Kernel methods, Random Matrix Theory, Information and complexity, Kernel methods for Machine Learning*

**2020 – 2021 Master 1 Applied Mathematics and Statistics; Toulouse School of Economics, France**

**Mathematics and Economic Decision** *Games and equilibria, Probability modelling, Strategic and dynamic optimisation, Time series, Martingales theory, Advanced functional analysis*

**2019 – 2022 Magister Econometrics and Statistics; Toulouse School of Economics, France**

**Economist Statistician** *Machine Learning, Stochastic simulation, Markov chains and applications, Data analysis, Programming, Statistics, Optimisation for big data*

**2017 – 2020 Double Degree Economics and Mathematics; Toulouse School of Economics, France**

**Economics, Mathematics and Computer Science applied to human and social sciences** *Statistics, Measure theory and probability, Algebra, Algorithmic, Micro and Macro Economics, Analysis and optimisation, Econometrics*

## WORK EXPERIENCE

**2022 Research; LIGM Paris, France**

*Research project on the feasibility and stability properties of the equilibria in Lotka Volterra models for theoretical ecology in large dimension. Study of the phase transition phenomena for the feasibility, Volterra Lyapunov stability and P-matrix properties in the context of random matrices and their asymptotic spectral properties.*

*Supervisor: Jamal Najim and Walid Hachem*

**2021 Data Scientist, Research & Development; 55 Paris, France**

*Research project in cognitive science, focused on decision-making processes concerning the acceptance of website cookies. The project included two experiments: AB-Tests to identify the impact of visual features on marketing indicators and a survey to evaluate the visual influence of consent notice on cognitive mechanisms.*

*Supervisor: Sabine Hamroun and Romain Warlop*

**2020 Data Scientist; EvidenceB Paris, France**

*Data Science project on a new interactive online learning tool using AI at the service of school curriculums. Data analysis, implementation of clustering and recommendation algorithms (collaborative filtering, spectral classification, k-means), creation of a Python package.*

*Supervisor: Xavier Dupré and Bruno Martin*

## PUBLICATIONS

**2025 False Discovery Estimation in Record Linkage; Statistics in Medicine**

*A methodology for estimating the False Discovery Proportion in Record Linkage using synthetic data.*

**2025 A Flexible Model for Record Linkage; Journal of the Royal Statistical Society Series C: Applied Statistics**

*A model for linking multiple data sources without access to unique identifiers.*

## TEACHING & SERVICE

**2025 – now CoMeEcon Seminar; VVSOR, The Netherlands**

*Biannual seminar, organised within the Mathematical Statistics section of the VVSOR, about computational statistics methods through time and developments. Role: organiser.*

**2024 - now Statistical Consultancy; Amsterdam UMC, The Netherlands**

*Statistical consultancy for early career medical researchers from the Amsterdam University Medical Centre. Role: consultant.*

**2023 - now Medical Research into Clinical Practice; Amsterdam UMC, The Netherlands**

*Leading exercises classes (working groups). Role: teaching assistant.*

**2023 – now Department outing committee; Amsterdam UMC, The Netherlands**

*Organisation of the annual outing for the Department of Epidemiology and Data Science. Role: organiser.*

## TALKS & POSTERS

**2025 Conference talk; International Conference on Statistics and Data Science, Sevilla, Spain**

*'Causal Record Linkage'*

**2025 Conference talk; International Biometric Society Channel Network Conference, Liège, Belgium**

*'False Discovery Estimation in Record Linkage'*

**2025 Conference poster; European Inference Meeting, Ghent, Belgium**

*'Causal Critical Issues and Novel Approaches to False Discovery in Causal Record Linkage'*

**2024 Conference talk; International Biometric Conference, Atlanta, United States**

*'Advances in Record Linkage'*

**2024 Conference talk; CompStats, Giessen, Germany**

*'A Stochastic EM approach to Record Linkage'*

**2023 Conference talk; International Biometric Society Channel Network Conference, Wageningen, The Netherlands**

*'Record Linkage: progress in practice'*

**2023 Seminar talk; Eerstelijnszorg, Public Health & Methodologie, Amsterdam UMC, The Netherlands**

*'Record Linkage'*

## OUTREACH

**2024 – now Fietsenwerkplaats Smerig; Amsterdam, The Netherlands**

*Volunteer work; build and repair bicycles; assemble and disassemble bikes, wheels, internal hub mechanisms.*

**2017 – 2022 Student support & Guidance; Toulouse, Paris, France**

*Educational support for children with scholar difficulties and for young students.*

## MISCELLANEOUS

Programming proficiency Python, R, Cpp, SAS, STATA and SQL
https://kayanerobach.github.io/
French (native), English (fluent), Spanish (intermediate), Dutch (intermediate)
Piano & Violin amateur player
Driving License (B)